



Time Series Analysis

Seminar 0: Introduction: components and exponential smoothing



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ **ACF**: Autocorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **GDP**: Gross Domestic Product
- ▣ **HICP**: Harmonised Index of Consumer Prices
- ▣ **INS**: Institutul Național de Statistică — the National Institute of Statistics (Romania)
- ▣ **MAE**: Mean Absolute Error
- ▣ **RMSE**: Root Mean Squared Error
- ▣ **STL**: Seasonal and Trend decomposition using Loess

Today's questions and route

- First seminar of the course, before the [Chapter 0](#) lecture: everything you need is on the next slides
- **Questions of the seminar**
 - ▶ how do we open the course notebooks and load a series?
 - ▶ which growth rate tells us how the economy is doing?
 - ▶ does today's value tell us anything about tomorrow's?
 - ▶ how good is the simplest possible forecast?
- **Part I:** setup, calculations on paper (A1–A3), the same ideas on real data (B1, B2)
- **Part II:** your turn (A4, B3, B4), a project idea (C1) and the critique of an AI answer (C2)
- Open the [seminar notebook](#) in Google Colab; nothing is handed in

Exercise map

Exercise	Question	Type	Model
A1	growth rates of a quarterly series	Solved	–
A2	a sample autocorrelation by hand	Solved	–
A3	naive forecasts and their errors	Solved	–
B1	Romanian GDP: which growth rate?	Solved	–
B2	EUR/RON: memory of levels and of changes	Solved	–
A4	the same calculations on new numbers	Proposed	A1–A3
B3	Romanian inflation: monthly and annual	Proposed	B1, B2
B4	electricity: naive or seasonal naive?	Proposed	A3
C1	a project idea	Open	–
C2	find the errors in an AI answer	Proposed	A1, A3, B1

- ▣ **[Solved]**: full solution in the slides and the notebook, a model to follow; **[Proposed]**: you solve it, the solution is discussed in class
- ▣ Part A: on paper; Part B: real data, each task ending with an interpretation question; Part C: open

Prerequisites for Today (1/3): series and growth rates

- **Time series** $y_1, \dots, y_T$: one variable observed at successive moments
 - ▶ t : the time index (quarter, month, day); T : the number of observations
 - ▶ frequency: quarterly ($m = 4$ quarters a year), monthly ($m = 12$), daily
 - ▶ unadjusted series contain **seasonality**: a pattern that repeats every m periods
- **Growth over one period**: $100 (y_t / y_{t-1} - 1)$, in %
 - ▶ y_{t-1} : the value of the previous period; the result is the change in percent of that value
 - ▶ for quarterly data: quarter on quarter (q/q)
- **Growth over a year**: $100 (y_t / y_{t-m} - 1)$, year on year (y/y)
 - ▶ y_{t-m} : the value of the same season one year earlier
 - ▶ compares the same season, so the seasonal pattern cancels out
- **Log difference**: $100 (\ln y_t - \ln y_{t-1}) \approx 100 (y_t / y_{t-1} - 1)$ for small changes
 - ▶ $\ln$: the natural logarithm; for a price, it is the **log return**

Prerequisites for Today (2/3): autocorrelation

- **Sample autocorrelation** at lag k : how strongly y_t is related to its own value k periods earlier

- ▶ $r_k = \frac{\sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}$, between -1 and 1

- ▶ $\bar{y}$: the sample mean; k : the lag, the number of periods between the two values compared

- ▶ numerator: the products of deviations from the mean, positive when y_t and y_{t-k} lie on the same side of the mean;
denominator: the sum of squared deviations

- **ACF** (autocorrelation function): $r_1, r_2, \dots$ plotted against k

- ▶ band $\pm 1.96/\sqrt{T}$: for independent noise, about 95% of the r_k fall inside it

- ▶ 1.96: the 97.5% quantile of the standard Normal distribution; a value outside the band is significant at 5%

Prerequisites for Today (3/3): naive forecasts and their errors

- **Naive forecast:** $\hat{y}_{T+h} = y_T$; **seasonal naive:** $\hat{y}_{T+h} = y_{T+h-m}$ ($h \leq m$)
 - ▶ $\hat{y}_{T+h}$: the forecast of the value h periods after the last observation T ; h : the horizon
 - ▶ naive: the last value; seasonal naive: the value of the same season one year earlier
- **Training set and test set:** split by time
 - ▶ forecasts are built on the training set and compared with the test set they have not seen
- **Errors** $e = y - \hat{y}$ on the test set
 - ▶ MAE (mean absolute error) = $\frac{1}{h} \sum |e|$: the average size of the errors, in the units of the series
 - ▶ RMSE (root mean squared error) = $\sqrt{\frac{1}{h} \sum e^2}$: penalises large errors more; always $\geq$ MAE
 - ▶ here h is the number of test values; the sums run over the test set

Data used today

Series	Source	Unit	Frequency	Exercise
Romanian real GDP	Eurostat, unadjusted	billion EUR, 2010 prices	quarterly	B1, C2
EUR/RON	BNR reference rate	lei per euro	daily, from 2015	B2
Romanian HICP	Eurostat	index, 2015 = 100	monthly, from 2015	B3
Electricity generation	Eurostat	TWh per month	monthly, from 2008	B4

- HICP: Harmonised Index of Consumer Prices; BNR: the National Bank of Romania; TWh: terawatt-hour
- Public sources, read online without a key; the notebook defines one function per source

Setup (1/2): open the notebook and save a copy

- ▣ **Question:** can you run Python on the course data in the browser?
- ▣ **Inputs:** Google Colab: Google's free notebook service; Python runs on Google's servers, a Google account is enough
- ▣ **Tasks**
 1. Open the [seminar notebook](#) in Colab.
 2. Choose *File* → *Save a copy in Drive*, so that your changes are kept.
 3. Run the cells of the section *Setup* in order with Shift + Enter; they import the libraries and define the data functions.
 4. If a cell fails, choose *Runtime* → *Restart session* and run again from the top.
- ▣ **Report:** nothing; you are ready when the check on the next slide prints the expected numbers

Setup (2/2): load a series

```
# Romanian real GDP, quarterly, unadjusted
g = read_eurostat('namq_10_gdp',
                  'Q.CLV10_MEUR.NSA.B1GQ.RO') / 1000
print(len(g), g.index[0].date(), g.index[-1].date())
print(g.tail(2).round(1))
g.plot()           # a first chart
```

- `read_eurostat(dataset, key)` reads one Eurostat series; the key lists the dimensions (frequency, unit, adjustment, indicator, country)

- **Expected output**

- ▶ 126 quarters, from 1995 Q1 to 2026 Q2
- ▶ last value: 45.88 billion EUR

- Eurostat revises its data: a slightly different last value is normal
- If the count differs a lot, check the key before going on

A1 [Solved]: Growth rates — task

- ▣ **Question:** did the Romanian economy collapse in the first quarter of 2025?
- ▣ **Inputs:** real GDP, billion EUR: 38.7, 45.6, 52.8, 59.2 (2024 Q1–Q4), 38.9, 46.8 (2025 Q1–Q2)
- ▣ **Tasks**
 1. Compute the quarter-on-quarter growth rates for 2024 Q2 to 2025 Q2.
 2. Compute the year-on-year growth rates for 2025 Q1 and 2025 Q2.
 3. Compute the log difference for 2025 Q1 and compare it with the quarter-on-quarter rate.
- ▣ **Report:** the two kinds of growth rates, and one sentence that answers the question
- ▣ **Notebook:** section A1

A1 [Solved]: Solution

Quarter	y_t	q/q, %	y/y, %
2024 Q1	38.7	–	–
2024 Q2	45.6	+17.8	–
2024 Q3	52.8	+15.8	–
2024 Q4	59.2	+12.1	–
2025 Q1	38.9	–34.3	+0.5
2025 Q2	46.8	+20.3	+2.6

□ q/q: quarter on quarter; y/y: year on year

□ **Step by step**

- ▶ 2025 Q1, q/q: $100 (38.9/59.2 - 1) = -34.3\%$
- ▶ 2025 Q1, y/y: $100 (38.9/38.7 - 1) = +0.5\%$
- ▶ log difference: $100 (\ln 38.9 - \ln 59.2) = -42.0\%$, far from -34.3% : the approximation fails for large changes

□ **Interpretation**

- ▶ no collapse: the fall of a third from Q4 to Q1 happens every winter; compared with a year earlier, GDP grew by 0.5%

A2 [Solved]: An autocorrelation by hand — task

- ▣ **Question:** is a high value usually followed by a high value in this short series?
- ▣ **Inputs:** $y = 2, 4, 3, 5, 4, 6$ ($T = 6$)
- ▣ **Tasks**
 1. Compute the mean $\bar{y}$ and the deviations $y_t - \bar{y}$.
 2. Compute the sum of squared deviations and the sum of products of consecutive deviations.
 3. Compute r_1 and r_2 , and compare them with the band $\pm 1.96/\sqrt{T}$.
- ▣ **Report:** $\bar{y}$, r_1 , r_2 , the band, and one sentence on what can be concluded from 6 observations
- ▣ **Notebook:** section A2

A2 [Solved]: Solution

t	y_t	$d_t = y_t - \bar{y}$	$d_t d_{t-1}$
1	2	-2	-
2	4	0	0
3	3	-1	0
4	5	1	-1
5	4	0	0
6	6	2	0
sum	24	0	-1

$$\bar{y} = 24/6 = 4; \sum d_t^2 = 4 + 0 + 1 + 1 + 0 + 4 = 10$$

$$r_1 = -1/10 = -0.10$$

$$\begin{aligned} \text{lag 2:} \\ \sum d_t d_{t-2} &= (-1)(-2) + (1)(0) + (0)(-1) + (2)(1) = 4, \\ \text{so } r_2 &= 0.40 \end{aligned}$$

$$\text{band: } \pm 1.96/\sqrt{6} = \pm 0.80$$

□ **Interpretation**

- both values lie well inside the band: 6 observations are far too few to detect memory

A3 [Solved]: Naive forecasts and their errors — task

- ▣ **Question:** for a seasonal series, which simple forecast is better: the last value or the same quarter last year?
- ▣ **Inputs:** quarterly sales; training set: 10, 14, 18, 12 (year 1) and 11, 15, 20, 13 (year 2); test set: 12, 16, 21, 14 (year 3)
- ▣ **Tasks**
 1. Write the naive forecasts and the seasonal naive forecasts ($m = 4$) for the four quarters of year 3.
 2. Compute the errors $e = y - \hat{y}$ of each method.
 3. Compute the MAE and the RMSE of each method.
- ▣ **Report:** a table with the forecasts and errors, the four error measures, and one sentence naming the better method
- ▣ Notebook: section A3

A3 [Solved]: Solution

	Q1	Q2	Q3	Q4
actual (year 3)	12	16	21	14
naive	13	13	13	13
error	-1	3	8	1
seasonal naive	11	15	20	13
error	1	1	1	1

□ **Naive:** every forecast is the last value, 13

► $MAE = (1 + 3 + 8 + 1)/4 = 3.25$; $RMSE = \sqrt{(1 + 9 + 64 + 1)/4} = 4.33$

□ **Seasonal naive:** each quarter of year 2

► $MAE = 1.00$; $RMSE = 1.00$

□ **Interpretation**

► the seasonal naive forecast is better: it keeps the seasonal pattern; the large Q3 error inflates the naive RMSE

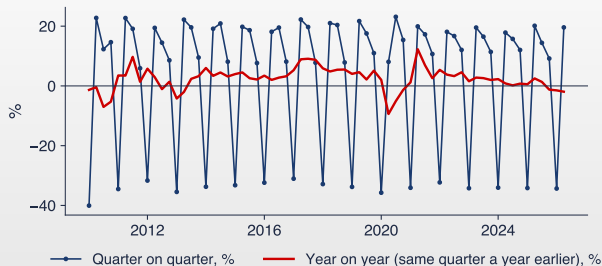
B1 [Solved]: Romanian GDP — Task

- **Question:** which growth rate should be reported for an unadjusted quarterly series?
- **Inputs:** Romanian real GDP, unadjusted, 1995 Q1 – 2026 Q2 (Eurostat)
- **Tasks**
 1. Compute the quarter-on-quarter and the year-on-year growth rates of the whole series.
 2. Plot the two growth rates since 2010 on the same chart.
 3. Count the negative values of each rate, and compute the average q/q growth of the first quarters.
 4. Interpretation: which of the two rates shows the recessions?
- **Report:** the counts of negative values, the average Q1 growth, the latest two rates, and the answer to the interpretation question
- **Notebook:** section B1

B1 [Solved]: Solution (1/2) — step by step

- **Quarter on quarter:** 125 values, 32 negative
 - ▶ the first quarters average -36.3% : a fall of a third every winter
 - ▶ standard deviation 24.6 percentage points: dominated by the season
- **Year on year:** 122 values, 28 negative
 - ▶ negative in the years 1997, 1998, 1999, 2000, 2001, 2009, 2010, 2012, 2013, 2020, 2025, 2026
 - ▶ average 2.9% , standard deviation 4.9 percentage points
- **Latest quarter, 2026 Q2**
 - ▶ q/q: $+19.6\%$ (spring after winter); y/y: -1.9%

B1 [Solved]: Solution (2/2) — chart and interpretation



- Blue: q/q swings between about -35% and $+20\%$ every year; red: y/y moves slowly, around the business cycle
- Interpretation:** only the y/y rate shows the recessions (2009–2010, 2020) and the slowdown of 2025–2026; for unadjusted data, report y/y growth

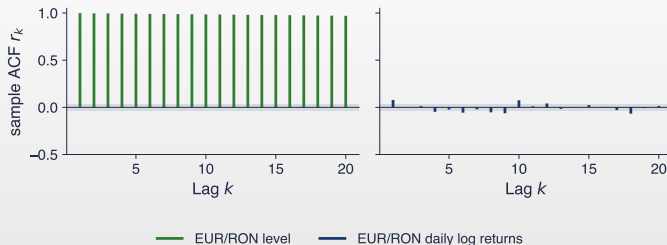
B2 [Solved]: EUR/RON: memory of levels and changes — task

- **Question:** does today's exchange rate help to forecast tomorrow's, and does today's change help to forecast tomorrow's change?
- **Inputs:** EUR/RON reference rate of the BNR, 5 January 2015 – 18 September 2026 (2 939 days)
- **Tasks**
 1. Compute the daily log returns $r_t = 100 (\ln P_t - \ln P_{t-1})$.
 2. Compute the sample ACF of the level and of the returns for lags 1–20.
 3. Compare r_1 of the returns with the band $\pm 1.96/\sqrt{T}$, and count the lags outside the band.
 4. Interpretation: is the naive forecast a sensible benchmark for the exchange rate?
- **Report:** r_1 and r_{20} of the level, r_1 and r_2 of the returns, the band, the number of lags outside it, and the answer to the interpretation question
- **Notebook:** section B2

B2 [Solved]: Solution (1/2) — step by step

- **Level:** $r_1 = 1.00$, $r_{20} = 0.97$
 - ▶ the level remembers itself for weeks: a slow decay, the signature of a trend
- **Returns:** standard deviation 0.13% a day; $r_1 = 0.078$, $r_2 = -0.0003$
 - ▶ band ± 0.036 ; 8 of the 20 lags lie outside it
 - ▶ r_1 is significant but small: yesterday's change explains less than 1% of the variance of today's change (r_1^2)

B2 [Solved]: Solution (2/2) — chart and interpretation



- Left: the ACF of the level stays close to 1; right: the ACF of the returns is almost flat; shaded: the band $\pm 1.96/\sqrt{T}$
- **Interpretation:** yes; the level of today is the best simple forecast of tomorrow's level, while past changes add very little: the naive forecast is the benchmark to beat

A4 [Proposed]: New numbers — task

- **Question:** can you repeat A1–A3 without the solution in front of you?
- **Inputs:** a quarterly series 120, 150, 132, 168, 126 (Q1 of year 1 to Q1 of year 2); a series 5, 7, 6, 8, 9; a series with $m = 3$: training 20, 30, 25, 22, 32, 27, test 23, 33, 29
- **Tasks**
 1. Model: A1, A2 and A3 [Solved].
 2. Compute the period-on-period growth rates, the log differences and the year-on-year rate of the last quarter.
 3. Compute r_1 of the series 5, 7, 6, 8, 9 and the band $\pm 1.96/\sqrt{T}$.
 4. Compute the MAE and the RMSE of the naive and seasonal naive forecasts of the test values.
- **Report:** the growth table, r_1 and the band, the four error measures, and one sentence naming the better forecast
- **Notebook:** section A4

B3 [Proposed]: Romanian inflation — task

- ▣ **Question:** does monthly inflation have a seasonal pattern that the annual rate hides?
- ▣ **Inputs:** Romanian HICP, 2015 = 100, monthly, 1 January 2015 – 1 August 2026 (Eurostat)
- ▣ **Tasks**
 1. Model: B1 and B2 [Solved].
 2. Compute the monthly (m/m) and the annual (y/y) inflation rates in %.
 3. Compute the average monthly inflation for each calendar month.
 4. Compute the ACF of the monthly rate for lags 1–24, and of the annual rate at lag 1.
 5. Interpretation: why is the annual rate so persistent?
- ▣ **Report:** the latest monthly and annual rates, the months with the highest and lowest average, r_1 and r_{12} of the monthly rate, r_1 of the annual rate, and the answer to the interpretation question
- ▣ **Notebook:** section B3

B4 [Proposed]: Electricity: naive or seasonal naive? — task

- ▣ **Question:** for monthly electricity generation, does the seasonal naive forecast beat the naive one?
- ▣ **Inputs:** net electricity generation in Romania, TWh per month (Eurostat); test set: August 2024 – July 2026 (24 months)
- ▣ **Tasks**
 1. Model: A3 [Solved].
 2. Split the series by time: the last 24 months form the test set.
 3. Forecast the test set with the naive and with the seasonal naive method ($m = 12$), using only the training set.
 4. Compute the MAE and the RMSE of each method, and plot both forecasts against the test data.
 5. Interpretation: why does the seasonal naive method not win clearly here?
- ▣ **Report:** the four error measures, the chart, and the answer to the interpretation question
- ▣ **Notebook:** section B4

C1: A project idea

- ▣ **Question:** how much of the variation of a Romanian monthly series is seasonal, and is the seasonal pattern changing?
- ▣ **Candidate series** (Eurostat or INS)
 - ▶ electricity generation, tourist arrivals, retail trade, industrial production
- ▣ **Steps**
 - ▶ plot the series and its year-on-year growth
 - ▶ compare the average of each calendar month in the first and in the last five years
 - ▶ after the lecture: decompose the series with STL and compare the seasonal amplitude over time
- ▣ A good first step for the team project of [Chapters 0](#) and [4](#)

C2 [Proposed]: Critique an AI answer — the answer

- **Prompt** sent to an AI assistant: “From the quarterly Romanian real GDP, compute the average annual growth rate and the RMSE of a naive forecast.”

```
g = read_eurostat('namq_10_gdp', 'Q.CLV10_MEUR.NSA.B1GQ.RO') / 1000
growth = g.diff() # annual growth rate, %
print('average annual growth:', growth.mean())
test = g.sample(frac=0.2, random_state=1).sort_index() # test set
train = g.drop(test.index) # training set
fc = train.reindex(g.index).ffill().shift(1).reindex(test.index) # naive
rmse = ((test - fc) ** 2).mean()
print('RMSE of the naive forecast:', rmse)
```

- **Conclusion of the AI:** “Romanian GDP grows by only 0.24% a year, and the naive forecast has an RMSE of 111.9 billion EUR, larger than GDP itself: forecasting GDP is hopeless.”

C2 [Proposed]: Critique an AI answer — task

- ▣ **Question:** can you trust code that runs without an error message?
- ▣ **Inputs:** the code and the conclusion on the previous slide; the definitions of the primer
- ▣ **Tasks**
 1. Model: A1, A3 and B1 [Solved].
 2. Find the three errors planted in the code.
 3. For each error, say how you would detect it: a check on the output, the primer, or a solved exercise.
 4. Write the corrected code, with the last 8 quarters as the test set, and report the corrected numbers.
 5. Say whether the conclusion “forecasting GDP is hopeless” survives.
- ▣ **Report:** one line per error in the format: request, answer, error, how found, correction; the corrected growth and RMSE
- ▣ Notebook: section C2

Practice at home

- **Finish the proposed exercises:** A4, B3, B4 and C2
 - ▶ each names its model, a solved exercise to follow
- **Try one more series**
 - ▶ repeat B1 for the euro area GDP (key Q.CLV10_MEUR.NSA.B1GQ.EA20) and compare the seasonal swings with Romania
- **Take the [Chapter 0](#) quiz** on the course website: 20 questions, each answer explained

Recap and next: the Chapter 0 lecture

□ Today

- ▶ for unadjusted data, report growth over a year, not over one quarter
- ▶ r_k : how much a series remembers its value k periods earlier
- ▶ levels with a trend remember a lot; daily changes of prices remember little
- ▶ the naive and seasonal naive forecasts are the benchmarks; split by time, then measure MAE and RMSE
- ▶ AI code can run and still be wrong

□ In the lecture

- ▶ How is the course organised and graded?
- ▶ Which components make up a time series?
- ▶ How does exponential smoothing forecast?
- ▶ Which forecasts win on real Romanian data?
- The seminar is practice and is not graded; the solutions of the [Proposed] tasks are discussed in class